STAMPose: Spatial-Temporal Attention-Mamba for Efficient 3D Human Pose Estimation

Hyun-il Kim[1] and Seung-bo Park[2]*

[1] Dept. of Advanced Material Processing & Engineering, Inha University, Incheon, Korea
akadjsam@inha.edu
[2] Dept. of Software Convergence Engineering, Inha University, Incheon, Korea
molaal@inha.ac.kr

Abstract. For monocular 2D-to-3D human pose lifting, recent purely Mamba-based methods exploit the linear complexity of state space models (SSMs) by applying them to both the spatial and temporal axes. At standard benchmark lengths (J=17 joints, T=243 frames), however, asymptotic complexity offers limited guidance for choosing each operator. The proposed STAMPose pairs spatial attention with a Dual-Branch Concatenation Temporal SSM (DC-TSSM) block, which fuses a bidirectional selective scan and a depthwise-convolution branch by channel concatenation. We select each operator from measured results: on the spatial axis, attention matches the accuracy of an SSM at 1.73× higher throughput, whereas on the temporal axis, DC-TSSM is more accurate than attention, which runs faster. On Human3.6M and MPI-INF-3DHP, STAMPose matches or surpasses prior SSM models while achieving 1.7–2.0× higher throughput than PoseMamba at comparable parameter counts. The source code is available at `https://github.com/akadjsam/STAMPose`.

Keywords: 3D Human Pose Estimation · State Space Models · Mamba · Attention · Hybrid Architectures

1 Introduction

3D human pose estimation (3D HPE) from monocular video is an important computer vision task with applications across virtual and augmented reality, sports analysis, and healthcare [1, 21, 24, 25]. Approaches to 3D pose estimation fall broadly into two categories: direct estimation, which regresses the 3D pose directly from the image, and 2D-to-3D lifting, which first detects the 2D keypoints and then maps them into 3D space. The lifting paradigm, which extracts 2D keypoints with a pre-trained 2D detector and lifts them to 3D, is particularly practical because it can be applied without expensive motion-capture equipment or multi-camera setups [13, 20, 28, 40]. Nevertheless, recovering accurate 3D positions from 2D keypoints remains challenging because of the depth ambiguity and self-occlusion inherent in monocular data.

* Corresponding author.

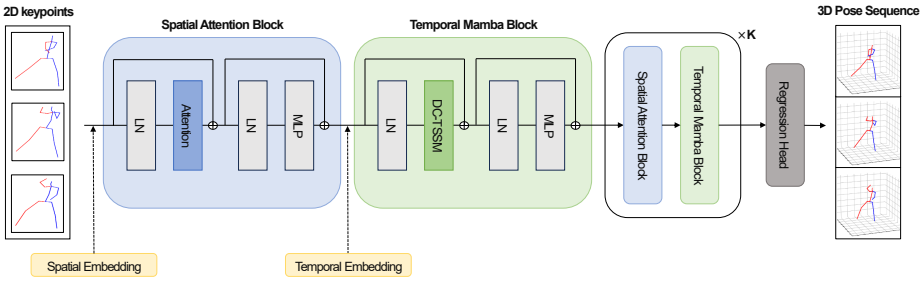


Fig. 1: Overall architecture of the proposed STAMPose framework, with alternating Spatial Attention Blocks and Temporal Mamba Blocks between the input embedding and the regression head.

Deep learning methods have been actively studied to address these difficulties. In particular, the introduction of the Transformer architecture has improved the estimation accuracy by modeling long-range dependencies [35]. PoseFormerV2 exploits the frequency domain to estimate the 3D pose from video sequences, while MixSTE proposes a Transformer-based seq2seq model with interleaved spatial and temporal blocks to capture cross-frame coherence [38, 41]. MotionBERT processes spatial and temporal information in two parallel streams and adaptively fuses them, forming a DSTformer [43]. Although these methods capture inter-joint relations and cross-frame temporal context well, the quadratic complexity of attention causes the computational cost to grow sharply with the input sequence length. MotionAGFormer strengthens spatial modeling by combining attention with a GCN in parallel while retaining temporal attention, whose cost grows quadratically with the sequence length [22].

State space model (SSM)-based architectures with linear complexity have gained traction to overcome this limitation [7, 8]. Mamba, which extends the structured state space model (S4), achieves both a global receptive field and linear complexity through a data-dependent selective scan mechanism [6]. SSM-based methods such as PoseMamba and SasMamba have applied this to 3D HPE, modeling spatio-temporal dependencies with linear complexity [5, 11]. However, purely SSM-based architectures flatten the joints into a fixed order for scanning when handling the non-Euclidean spatial topology of 3D skeletal data because they are tailored to one-dimensional sequence processing. Bidirectional scanning and structure-aware scanning have been proposed to mitigate this, but such schemes capture inter-joint interactions only indirectly, through the scan order. This is fundamentally different from attention, which compares all joint pairs directly, regardless of order.

In this paper, we examine, through ablation studies, how this mechanistic difference affects accuracy and throughput (Sec. 4.6), and accordingly propose STAMPose (Spatial-Temporal Attention-Mamba Pose estimation), a hybrid architecture that assigns different operators to the spatial and temporal axes, as shown in Fig. 1. At the sequence lengths we consider, asymptotic complexity

offers limited guidance for this choice: the spatial axis (J=17) is short enough for the quadratic cost of attention to remain small, and attention is in fact faster along the temporal axis (T=243). We therefore select the operator on each axis from the accuracy and throughput measured in our ablations, and STAMPose deploys attention on the spatial axis and the proposed Dual-Branch Concatenation Temporal SSM (DC-TSSM) on the temporal axis.

Our contributions are as follows:

1. **Axis-wise operator selection from measured results.** Swapping one operator at a time at J=17 and T=243, we find that spatial attention ties with a spatial SSM in accuracy (40.8 mm) at 1.73× higher throughput, whereas temporal attention is faster but less accurate than DC-TSSM (42.7 vs. 40.8 mm); we therefore select the spatial operator by throughput and the temporal one by accuracy.
2. **A DC-TSSM temporal block for improved accuracy.** In place of the element-wise multiplicative gating of the standard Mamba block, we propose DC-TSSM, which combines a bidirectional selective-scan branch with an independent depthwise-convolution branch through channel concatenation. Applying the time-axis scan restriction and the dual-branch design together reduces P1 by 1.0 mm relative to the standard Mamba block with spatial-temporal scanning at the same parameter count, a larger gain than either choice alone yields (0.1 mm each).
3. **A favorable accuracy–throughput trade-off.** The resulting configuration maintains accuracy on par with or better than that of purely SSM-based models while achieving 1.7–2.0× higher throughput at comparable parameter counts.

2 Related Work

2.1 3D Human Pose Estimation

Video-based 3D human pose estimation is divided into multi-view and monocular approaches [13, 28]. Recent research has concentrated on the monocular setting because multi-view methods require multiple cameras. Among the monocular approaches, 2D-to-3D lifting has become mainstream in terms of efficiency and accuracy because it focuses on geometric mapping and has evolved to exploit the temporal context of multiple frames to alleviate depth ambiguity and occlusion in a single frame [20, 27, 28, 34, 40].

2.2 Transformer-Based Methods

As the Transformer, successful in natural language processing, was introduced to vision tasks, the field of 3D HPE likewise adopted the attention mechanism to model spatio-temporal dependencies [3, 15–17, 29, 30, 32, 33, 35, 37, 38, 41, 43]. However, the quadratic complexity and memory requirements of attention act as a deployment constraint for 3D HPE models that process long input sequences.

2.3 State Space Models

State space models (SSMs) have emerged as an efficient alternative for overcoming the computational limits of the Transformer [7,8]. An SSM is a linear system that maps an input $x(t)$ to a hidden state $h(t)$ and an output $y(t)$, defined by the following continuous-time ordinary differential equations:

$$h'(t) = Ah(t) + Bx(t), \tag{1}$$

$$y(t) = Ch(t). \tag{2}$$

Here $A \in \mathbb{R}^{N\times N}$ is the state transition matrix, and $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{1\times N}$ are the projection parameters. To apply this continuous-time model within a deep learning architecture, it is discretized using the Zero-Order Hold (ZOH) scheme:

$$\bar{A} = \exp(\Delta A), \tag{3}$$

$$\bar{B} = (\Delta A)^{-1}\big(\exp(\Delta A) - I\big)\Delta B. \tag{4}$$

The discrete-time model is then computed by the recurrence:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \tag{5}$$

$$y_t = Ch_t. \tag{6}$$

Conventional SSMs cannot select information because A, B, and C are fixed independently of the input. Mamba improves on this by introducing a data-dependent selective scan mechanism that discards irrelevant information and retains meaningful long-range context, achieving linear complexity $O(L)$ in the sequence length [6]. Subsequently, VMamba, Vision Mamba, and MambaVision adapted Mamba to 2D image processing [9, 18, 42].

SSM-based methods have also been applied to 3D HPE. PoseMamba proposes a bidirectional global-local spatio-temporal SSM that operates with linear complexity in place of attention, while SasMamba proposes SAS-SSM, which combines structure-aware convolution with stride scanning to preserve human spatial topology and capture multi-scale spatio-temporal dependencies [5, 11].

Hybrid approaches have also been proposed to address the spatial-modeling limitations of pure SSMs. Pose Magic is an attention-free hybrid that combines spatial and temporal Mamba blocks with a GCN capturing neighboring-joint relations and fuses the two representations adaptively; HGMamba likewise models spatial topology with graph-based components, and VIMCAN uses cross-attention for RGB-IMU fusion [4,36,39]. StructMamPose retains a Mamba backbone, anchoring its state updates with spatio-temporal identities to preserve valid skeletal dependencies and adding a view-transformation module to improve depth reasoning [14]. These methods strengthen SSM-based spatial modeling with graph priors, structural anchoring, or cross-modal fusion, whereas STAMPose compares joint pairs directly with spatial self-attention and concentrates its SSM design on the temporal axis; Sec. 4.6 reports the accuracy and throughput of this choice at J=17.

3 Method

3.1 Network Architecture

The overall architecture of the proposed STAMPose is shown in Fig. 1. STAMPose takes as input a 2D keypoint sequence $\mathbf{x} \in \mathbb{R}^{T\times J\times 2}$ representing J joints over T frames, and outputs a 3D keypoint sequence $\mathbf{y} \in \mathbb{R}^{T\times J\times 3}$. After projecting the input into a high-dimensional feature space through an initial embedding layer, we add spatial and temporal positional embeddings that encode per-joint and per-frame information. The features then pass through K pairs of a Spatial Attention Block (S) and a Temporal Mamba Block ($\mathcal{T}$), stacked in alternation as $(S\mathcal{T})^K$ for $2K$ blocks in total. Both block types use the same residual structure of LayerNorm–token mixer–LayerNorm–MLP, where the token mixer is attention for S and DC-TSSM for $\mathcal{T}$. The alternating order and the scan strategy are validated through the ablation studies in Sec. 4.6.

3.2 Spatial Attention Block

Because Sec. 4.6 shows that the two operators achieve comparable accuracy along the spatial axis at J=17, we implement the spatial operation with attention, which offers higher GPU-parallelization efficiency:

$$x_s = \mathrm{Reshape}(x,\, [B{\cdot}T,\, J,\, C]), \tag{7}$$

$$z_s = \mathrm{Attn}(\mathrm{LN}(x_s)) + x_s. \tag{8}$$

The $O(J^2)$ complexity incurred here has a limited effect on the overall computation of the network because the number of joints J is relatively small.

3.3 Temporal Mamba Block

In the temporal-context learning stage, we place DC-TSSM in the token-mixer position instead of attention to improve temporal-modeling accuracy; its structure is shown in Fig. 2.

DC-TSSM is reconfigured for the temporal modeling of 3D HPE, drawing inspiration from the MambaVision mixer [9]. DC-TSSM delegates spatial topology modeling to the preceding Spatial Attention Block and extracts temporal dependencies through a bidirectional selective scan by restricting the scan target to the temporal axis (T). After an initial linear projection, the input tensor is split along the channel dimension into a main and an auxiliary branch. Unlike the standard Mamba block, whose auxiliary branch acts as an element-wise multiplicative gate, DC-TSSM places an independent 1D depthwise convolution in the auxiliary branch and combines the two branches by channel concatenation, so that the long-range features of the scan branch and the local temporal features of the convolution branch enter the output projection as separate channels.

Main Branch (SSM). Along the temporal axis, the input passes through a 1D depthwise convolution (DWConv) and a SiLU activation to perform temporal

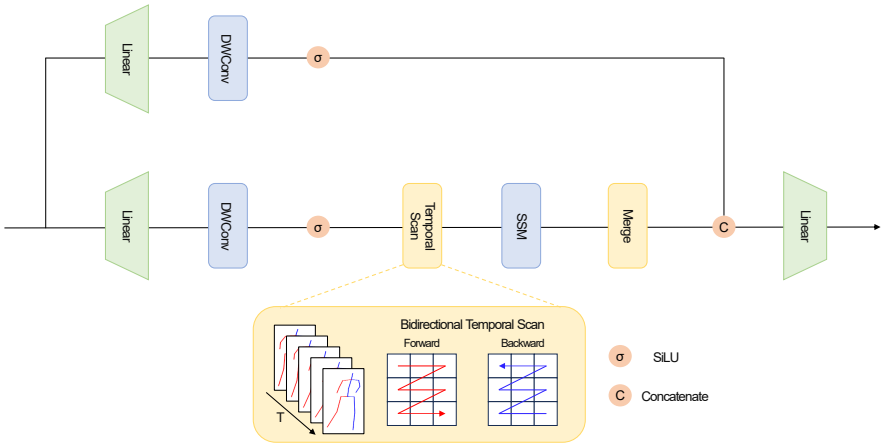


Fig. 2: Structure of the DC-TSSM block: a bidirectional-scan main branch and a depthwise-convolution auxiliary branch fused by channel concatenation.

smoothing, after which forward and backward selective scans extract long-range temporal dependencies with linear complexity $O(T)$:

$$z_{\text{ssm}}^{\text{fwd}} = \text{SSM}\big(\text{SiLU}(\text{DWConv}(x_{\text{main}}))\big), \tag{9}$$

$$z_{\text{ssm}}^{\text{bwd}} = \text{SSM}\big(\text{Flip}(\text{SiLU}(\text{DWConv}(x_{\text{main}})))\big), \tag{10}$$

$$z_{\text{ssm}} = z_{\text{ssm}}^{\text{fwd}} + \text{Flip}(z_{\text{ssm}}^{\text{bwd}}). \tag{11}$$

Auxiliary Branch. The auxiliary branch learns local temporal features through an independent 1D depthwise convolution and a SiLU activation:

$$z_{\text{aux}} = \text{SiLU}(\text{DWConv}(x_{\text{aux}})). \tag{12}$$

Finally, the two branch outputs are concatenated along the channel dimension and linearly projected:

$$z_{\text{temporal}} = \text{Linear}(\text{Concat}(z_{\text{ssm}},\, z_{\text{aux}})). \tag{13}$$

3.4 Loss Function

The proposed network is trained in an end-to-end manner, and its loss formulation follows previous work [43]. The final loss function $\mathcal{L}$ is defined as:

$$\mathcal{L} = \mathcal{L}_{3D} + \alpha\mathcal{L}_t + \beta\mathcal{L}_m. \tag{14}$$

Here $\mathcal{L}_{3D}$ is the Mean Per-Joint Position Error (MPJPE) loss between the predicted 3D pose and the ground truth. $\mathcal{L}_t$ is the Temporal Consistency Loss (TCLoss) for generating natural poses, and $\mathcal{L}_m$ is the Mean Per-Joint Velocity Error (MPJVE) loss for improving temporal velocity consistency [10, 28]. α and β are the weights controlling the contribution of each loss term, set to α=0.5 and β=20.

Table 1: STAMPose model variants. K is the number of alternating Spatial-Temporal block pairs, and d_m is the embedding dimension.

Variant	K (ST pairs)	Total blocks	d_m	T	Params	MACs
STAMPose-S	10	20	64	243	0.8M	3.5G
STAMPose-B	10	20	128	243	3.3M	13.8G
STAMPose-L	20	40	128	243	6.7M	27.5G

4 Experiments

We evaluate the performance and computational efficiency of STAMPose on the Human3.6M and MPI-INF-3DHP datasets and further report a zero-shot evaluation on 3DPW [12, 19, 23].

4.1 Datasets and Evaluation Metrics

Human3.6M is a representative 3D HPE dataset comprising approximately 3.6 million video frames in which 11 subjects perform 15 daily actions in an indoor environment [12]. Following the standard protocol, we use S1, S5, S6, S7, and S8 for training and S9 and S11 for testing. As evaluation metrics, we use Protocol 1 (P1), which measures MPJPE after root-joint alignment, and Protocol 2 (P2, P-MPJPE), which measures the error after Procrustes-based alignment; both are reported in mm.

MPI-INF-3DHP is a dataset collected with a motion-capture system in studio (green-screen and non-green-screen) and outdoor environments, covering diverse backgrounds and dynamic motions [23]. This dataset is well suited to evaluating the generalization performance of a model because it contains more complex conditions than Human3.6M. We report PCK, AUC, and MPJPE.

4.2 Implementation Details

Model Variants. We construct three model variants by adjusting the network depth and the embedding dimension: STAMPose-S (Small), STAMPose-B (Base), and STAMPose-L (Large); their detailed specifications are given in Tab. 1. The expansion ratio of the MLP branch is fixed at 2 in all models.

Training Setup. The model was implemented and trained using the PyTorch framework on a single NVIDIA RTX 3090 Ti GPU. Following previous work, we used as input either the detection results of the Stacked Hourglass network or ground-truth 2D keypoints, and applied horizontal-flipping data augmentation [5, 11, 26, 43]. The batch size was set to 4, and the AdamW optimizer was used. Training was carried out for 120 epochs, with an exponential decay schedule of factor 0.99 from an initial learning rate of 2×10^{-4}. The weight decay was set to 0.01.

Table 2: Quantitative comparisons on the Human3.6M dataset. T denotes the number of input frames, and * marks methods that estimate only the center frame. P1 and P2 denote MPJPE and P-MPJPE (mm), respectively, and P1† denotes P1 with 2D ground-truth keypoints as input. In all tables, the best result is shown in **bold**, and the second-best result is underlined; tied results share the same mark.

Method	T	Params	MACs	MACs/frame	P1	P2	P1†
MixSTE [38]	243	33.6M	139.0G	572M	40.9	32.6	21.6
P-STMO* [32]	243	6.2M	0.7G	3M	42.8	34.4	29.3
StridedFormer* [16]	351	4.0M	0.8G	2M	43.7	35.2	28.5
STCFormer [33]	243	4.7M	19.6G	80M	41.0	32.0	21.3
PoseFormerV2* [41]	243	14.4M	4.8G	20M	45.2	35.6	–
GLA-GCN* [37]	243	1.3M	1.5G	6M	44.4	34.8	21.0
MotionBERT [43]	243	42.3M	174.8G	719M	39.2	32.9	17.8
HDFormer [3]	96	3.7M	0.6G	6M	42.6	33.1	21.6
MotionAGFormer-L [22]	243	19.0M	78.3G	322M	38.4	32.5	17.3
KTPFormer [29]	243	33.7M	69.5G	286M	40.1	<u>31.9</u>	19.0
TCPFormer [17]	243	35.1M	109.2G	449M	**37.9**	**31.7**	15.5
HGMamba [4]	243	14.2M	64.5G	265M	38.7	32.9	<u>13.2</u>
PoseMamba-S [11]	243	0.9M	3.6G	15M	41.8	35.0	20.0
PoseMamba-B [11]	243	3.4M	13.9G	57M	40.8	34.3	16.8
PoseMamba-L [11]	243	6.7M	27.9G	115M	<u>38.1</u>	32.5	15.6
SasMamba [5]	243	0.64M	1.3G	5M	41.5	34.8	21.4
SasMamba-large [5]	243	4.1M	8.56G	35M	39.8	33.6	20.9
STAMPose-S	243	0.8M	3.5G	14M	40.8	34.3	18.6
STAMPose-B	243	3.3M	13.8G	57M	38.7	33.0	14.0
STAMPose-L	243	6.7M	27.5G	113M	**37.9**	32.2	**12.3**

4.3 Performance Comparison on Human3.6M

In this section, we compare STAMPose with existing state-of-the-art architectures on the Human3.6M benchmark. To assess the structural effect objectively, Table 2 includes only models trained without pre-training on external datasets.

The experimental results show that STAMPose achieves lower P1 than PoseMamba at the S and B scales and comparable P1 at the L scale. In particular, STAMPose-B (3.3M parameters) attains a P1 error of 38.7 mm, 1.1 mm lower than the larger SasMamba-large (39.8 mm), while STAMPose-L reaches 37.9 mm, on par with TCPFormer (37.9 mm) with about one-fifth of its parameters and competitive with PoseMamba-L (38.1 mm) and MotionAGFormer-L (38.4 mm). Training STAMPose-L with three seeds yields 37.9, 37.8, and 38.1 mm (37.9 ± 0.2 mm, sample SD); since only a single run is reported for PoseMamba-L (38.1 mm), we regard the detected-input accuracy of the two models as comparable.

Accuracy Analysis by Input Quality. With ground-truth (GT) input (P1†), STAMPose-L shows the lowest error among the compared models at 12.3 mm,

Table 3: Inference efficiency comparison. All FPS values are measured by the authors on a single NVIDIA RTX 3090 Ti (batch size 32, T=243, FP32, random synthetic inputs, averaged over 100 runs after warm-up), using the officially released code and checkpoints for the baselines. Baseline parameter counts and MACs are taken from their original papers. Speedup (relative to PoseMamba) and the bold/underline marks are computed within each scale group.

Method	Params	MACs/frame	P1 ↓	FPS ↑	Speedup
SasMamba	0.64M	5M	<u>41.5</u>	<u>36,017</u>	–
PoseMamba-S	0.9M	15M	41.8	24,263	1.00×
STAMPose-S	0.8M	14M	**40.8**	**41,675**	1.72×
SasMamba-large	4.1M	35M	<u>39.8</u>	9,051	–
PoseMamba-B	3.4M	57M	40.8	<u>12,692</u>	1.00×
STAMPose-B	3.3M	57M	**38.7**	**25,058**	1.97×
PoseMamba-L	6.7M	115M	<u>38.1</u>	<u>6,312</u>	1.00×
STAMPose-L	6.7M	113M	**37.9**	**12,469**	1.98×

3.3 mm lower than PoseMamba-L (15.6 mm) and 0.9 mm lower than the runner-up HGMamba (13.2 mm), whereas the gap from PoseMamba-L is 0.2 mm with detection input (P1). Because P1 and P1† come from separately trained models, we treat the two gaps as separate observations (Sec. 4.8). The main advantage of STAMPose lies in the inference efficiency discussed next.

Inference Efficiency Comparison. The theoretical operation count (MACs) does not always reflect the actual inference throughput. Hence, we also report the throughput measured under identical conditions in Tab. 3.

The measurements show that STAMPose achieves 1.7–2.0× higher throughput than the comparably sized PoseMamba at every scale, while its P1 is comparable or lower. Because the two models differ in both the spatial operator and the temporal-block configuration, we locate the source of this gain with the ablation in Tab. 5, which fixes the temporal block to DC-TSSM and swaps only the spatial operator. In this ablation, the fully SSM variant runs at a throughput similar to that of PoseMamba-S (24,097 vs. 24,263 FPS), and replacing its spatial SSM with attention raises the throughput by a factor of 1.73, close to the 1.72× speedup of STAMPose-S over PoseMamba-S. This indicates that, at the S scale, replacing the sequential scan along the spatial axis with attention accounts for most of the throughput gain.

SasMamba-large shows lower throughput than PoseMamba-B in the same group despite having fewer MACs/frame. This suggests that the structure-aware convolution and stride scanning of SasMamba execute fewer MACs per second on the GPU than the selective scans of PoseMamba.

Table 4: Quantitative comparisons on MPI-INF-3DHP. Prior results are taken from their original papers. '–' indicates that PCK/AUC were not reported by PoseMamba and SasMamba.

Method	T	**PCK** ↑	**AUC** ↑	**MPJPE** ↓
MixSTE [38]	27	94.4	66.5	54.9
P-STMO [32]	81	97.9	75.8	32.2
STCFormer [33]	81	98.7	83.9	23.1
PoseFormerV2 [41]	81	97.9	78.8	27.8
GLA-GCN [37]	81	98.5	79.1	27.8
HDFormer [3]	96	98.7	72.9	37.2
MotionAGFormer-L [22]	81	98.2	85.3	16.2
KTPFormer [29]	81	98.9	85.9	16.7
TCPFormer [17]	81	<u>99.0</u>	<u>87.7</u>	15.0
PoseMamba [11]	81	–	–	<u>14.5</u>
SasMamba [5]	81	–	–	18.0
STAMPose	81	**99.6**	**89.2**	**14.2**

4.4 Performance Comparison on MPI-INF-3DHP

We conduct experiments on the MPI-INF-3DHP dataset, which includes real-world scenarios, to evaluate the robustness of STAMPose. For a fair comparison, we follow previous work [5, 11, 22]. We use 81 input frames on the test set, and Tab. 4 compares the results with existing methods.

At T=81, STAMPose records 14.2 mm, the lowest MPJPE among the compared methods. This is 0.3 mm lower than PoseMamba (14.5 mm) and surpasses the attention–GCN hybrid MotionAGFormer-L (16.2 mm) and the purely Mamba-based SasMamba (18.0 mm). In addition, STAMPose records 99.6% and 89.2% on the PCK and AUC metrics, respectively, showing competitive performance relative to existing methods.

4.5 Zero-Shot Evaluation on 3DPW

We evaluate Human3.6M-trained STAMPose-B, PoseMamba-B, and SasMamba-large checkpoints on the test split of 3DPW, an in-the-wild dataset captured with IMUs and a moving camera, without using 3DPW for training, fine-tuning, or checkpoint selection [19]. Under our custom scale-aligned 14-joint protocol, STAMPose-B obtains a scale-normalized MPJPE (N-MPJPE) of 146.32 mm versus 168.80 mm for PoseMamba-B with the OpenPose detections provided with 3DPW, and 97.86 mm versus 110.48 mm with projected ground-truth 2D inputs [2, 31]. These values are meant for comparison within our protocol rather than with standard SMPL-based 3DPW results. As single-checkpoint results, they are preliminary; attributing the differences to camera motion, occlusion, or specific components requires controlled experiments. Evaluation details and the PA-MPJPE and N-MPJVE results are given in Sec. B of the supplementary material.

Table 5: Ablation study on the spatial/temporal operator assignment. Starting from the proposed configuration (Spatial Attention + DC-TSSM), we swap one operator at a time: the spatial operator (row 1 vs. row 3) and the temporal operator (row 2 vs. row 3). The temporal SSM denotes the proposed DC-TSSM block, and the spatial SSM adopts the spatial SSM block of PoseMamba with its original scan strategy [11]. P1 is MPJPE under the protocol of Tab. 2. FPS is the inference throughput measured under the protocol in Tab. 3. Speedup is relative to the fully SSM variant.

Spatial	**Temporal**	**Params**	**P1** ↓	**FPS** ↑	**Speedup**
SSM	SSM	0.8M	**40.8**	24,097	1.00×
Attn	Attn	0.7M	<u>42.7</u>	**46,930**	1.95×
Attn	SSM	0.8M	**40.8**	<u>41,675</u>	1.73×

Table 6: Ablation study on the scan strategy and architecture of the Temporal Mamba Block. For scan strategies, 'Temporal-Only' scans the time axis, whereas 'Spatial-Temporal' scans both the joint and time axes, so that spatial mixing is performed by both the scan and the preceding Spatial Attention Block. All variants use the identical Spatial Attention Block.

Scan Strategy	**Architecture**	T	**Params**	**P1** ↓
Spatial-Temporal	Standard Mamba	243	0.8M	41.8
Spatial-Temporal	DC-TSSM	243	0.8M	<u>41.7</u>
Temporal-Only	Standard Mamba	243	0.8M	<u>41.7</u>
Temporal-Only	DC-TSSM	243	0.8M	**40.8**

4.6 Ablation Studies

In this section, we analyze how each design element of STAMPose affects the performance on the Human3.6M dataset. All experiments were conducted on the S model.

Ablation Study on the Operator Assignment. In Tab. 5, replacing the spatial operator with an SSM leaves P1 unchanged at 40.8 mm while lowering the throughput by a factor of 1.73, so at J=17 we select the spatial operator by throughput. By contrast, replacing the temporal operator with attention yields the highest throughput (46,930 FPS) but worsens P1 to 42.7 mm, higher than even PoseMamba-S (41.8 mm), so at T=243 we select the temporal operator by accuracy.

Analysis of the Temporal Block Design. In Tab. 6, restricting the scan to the time axis under the Standard Mamba structure yields a 0.1 mm improvement. The gain of the DC-TSSM structure depends on the scan condition: it is 0.1 mm under the Spatial-Temporal condition and grows to 0.9 mm under the Temporal-Only condition, where DC-TSSM reaches 40.8 mm versus 41.7 mm for

Table 7: Ablation study on the block arrangement strategy. S and $\mathcal{T}$ denote the Spatial Attention Block and the Temporal Mamba Block, respectively (K=10, $2K$=20 blocks in total).

Description	Arrangement Pattern	Params	P1 ↓
Grouped (Temporal-first)	$\mathcal{T}^{10}S^{10}$	0.8M	46.5
Grouped (Spatial-first)	$S^{10}\mathcal{T}^{10}$	0.8M	42.8
Alternating (Temporal-first)	$(\mathcal{T}S)^{10}$	0.8M	<u>42.2</u>
Alternating (Spatial-first)	$(S\mathcal{T})^{10}$	0.8M	**40.8**

Standard Mamba. Whereas Standard Mamba uses an auxiliary path for multiplicative gating, DC-TSSM applies an independent depthwise convolution to the auxiliary path and combines it with the scan branch by channel concatenation. Because the auxiliary-branch operation and the fusion rule change together, this gain reflects their joint effect. One plausible explanation is that the local temporal features of the auxiliary depthwise convolution complement the long-range dependencies of the main branch; separating the contributions of the auxiliary-branch operation and the fusion rule requires the branch-isolation ablation noted in Sec. 4.8. Finally, the configuration that applies both choices (Temporal-Only scan + DC-TSSM) achieves 40.8 mm, a 1.0 mm improvement over 41.8 mm that exceeds the sum of the individual gains (0.1 mm each); this non-additive pattern suggests an interaction between the two choices at the reported precision. We adopt this configuration, which yields the lowest P1 in Tab. 6.

Analysis of the Hybrid Block Arrangement Strategy. Table 7 shows the performance change according to the arrangement order of the Spatial Attention (S) and Temporal Mamba ($\mathcal{T}$) blocks, comparing grouped and alternating arrangements based on a total of 20 blocks (K=10).

The grouped patterns (46.5 and 42.8 mm) consistently show higher error than the alternating patterns (42.2 and 40.8 mm). In 2D image classification, a grouped arrangement (local feature extraction in the early layers followed by global context integration in later layers) has been reported to be advantageous [9]. In our 3D HPE setting, the alternating arrangements perform better. A possible reason is that grouping restricts the interaction between the spatial and temporal axes to the single transition between the two groups, whereas alternation allows the two axes to interact in every block pair. Among the alternating patterns, the spatial-first $(S\mathcal{T})^{10}$ records a 1.4 mm lower error than the temporal-first $(\mathcal{T}S)^{10}$. Since the two alternating patterns differ only in whether the stack begins with S or $\mathcal{T}$ (and correspondingly ends with $\mathcal{T}$ or S), one plausible explanation is that the spatial-first order establishes the per-frame joint structure before any temporal mixing takes place.

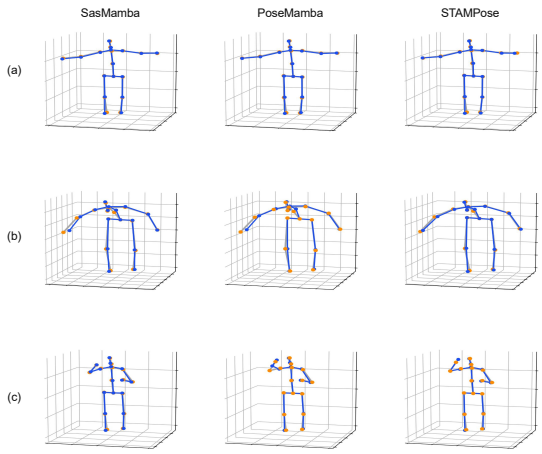


Fig. 3: Qualitative comparison with SasMamba-large and PoseMamba-B on Human3.6M. The gray skeleton is the ground-truth 3D pose. The blue skeleton is the estimated 3D pose.

4.7 Qualitative Analysis

For a parameter-controlled qualitative comparison, we compare STAMPose-B (3.3M) with PoseMamba-B (3.4M) and SasMamba-large (4.1M), which have comparable parameter scales. Figure 3 presents qualitative results randomly selected from the evaluation set of the Human3.6M dataset [12]. In these examples, STAMPose produces visually more plausible 3D poses than the purely Mamba-based PoseMamba and SasMamba [5, 11].

Figure 4 extends the comparison to in-the-wild videos. Here, STAMPose produces plausible poses under camera and background variations where the purely Mamba-based methods fail (red arrows).

4.8 Limitations

This study has the following limitations. First, accuracy was evaluated at T=243 (Human3.6M) and T=81 (MPI-INF-3DHP), and throughput at T=243. Since the Human3.6M checkpoints use a learned temporal positional embedding of length 243, longer inputs require extending or interpolating it, or retraining; memory and accuracy scaling at longer sequences thus remain to be verified despite the linear complexity of the selective scan. Second, the ablations supporting our operator selection and DC-TSSM design are reported for the S model at J=17 and T=243; controlled-noise and branch-isolation ablations, which would separate detector-noise and auxiliary-branch effects, are left to future work together with denoising or auxiliary-branch variants. Third, the accuracy advantage of STAMPose is most evident with ground-truth 2D input and narrows with detection input (Tab. 2), whereas the efficiency advantage is independent

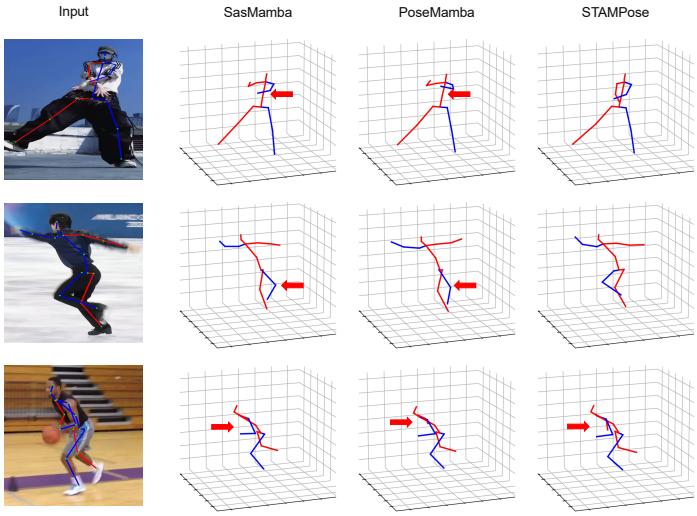


Fig. 4: Qualitative comparison with SasMamba-large and PoseMamba-B on challenging in-the-wild videos. The red arrows indicate incorrect estimations.

of input quality. Improving robustness to imperfect 2D keypoints, for example by coupling our method with a more accurate 2D detector, remains a subject for future work.

5 Conclusions

We presented STAMPose, which places attention on the spatial axis and DC-TSSM on the temporal axis for 2D-to-3D lifting, and reported its accuracy–throughput trade-off. STAMPose yields 1.7–2.0× higher throughput than PoseMamba at comparable or better accuracy. In our ablations at J=17 and T=243, the two spatial operators tie in accuracy while attention achieves higher throughput, and DC-TSSM is more accurate than temporal attention; these measurements support our operator selection for this setting. Extending this analysis to longer sequences and to controlled input perturbations is a natural next step.

References

1. Baumgartner, T., Klatt, S.: Monocular 3D human pose estimation for sports broadcasts using partial sports field registration. In: IEEE Conf. Comput. Vis. Pattern Recog. Worksh. pp. 5109–5118 (2023). https://doi.org/10.1109/CVPRW59228.2023.00539
2. Cao, Z., Hidalgo, G., Simon, T., Wei, S.E., Sheikh, Y.: OpenPose: Realtime multi-person 2D pose estimation using part affinity fields. IEEE Trans. Pattern Anal. Mach. Intell. **43**(1), 172–186 (2021). https://doi.org/10.1109/TPAMI.2019.2929257
3. Chen, H., He, J.Y., Xiang, W., Cheng, Z.Q., Liu, W., Liu, H., Luo, B., Geng, Y., Xie, X.: HDFormer: High-order directed Transformer for 3D human pose estimation. In: IJCAI. pp. 581–589 (2023). https://doi.org/10.24963/ijcai.2023/65
4. Cui, H., Hayama, T.: HGMamba: Enhancing 3D human pose estimation with a HyperGCN-Mamba network. In: Int. Joint Conf. Neural Netw. pp. 1–8 (2025). https://doi.org/10.1109/IJCNN64981.2025.11227716
5. Cui, H., Hua, W., Huang, R., Jia, S., Hayama, T.: SasMamba: A lightweight structure-aware stride state space model for 3D human pose estimation. In: IEEE/CVF Winter Conf. Appl. Comput. Vis. pp. 2721–2730 (2026). https://doi.org/10.1109/WACV61042.2026.00267
6. Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. In: Conf. Lang. Model. (COLM) (2024)
7. Gu, A., Goel, K., Ré, C.: Efficiently modeling long sequences with structured state spaces. In: Int. Conf. Learn. Represent. (2022)
8. Gu, A., Gupta, A., Goel, K., Ré, C.: On the parameterization and initialization of diagonal state space models. In: Adv. Neural Inform. Process. Syst. pp. 35971–35983 (2022). https://doi.org/10.52202/068431-2607
9. Hatamizadeh, A., Kautz, J.: MambaVision: A hybrid Mamba-Transformer vision backbone. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 25261–25270 (2025). https://doi.org/10.1109/CVPR52734.2025.02352
10. Hossain, M.R.I., Little, J.J.: Exploiting temporal information for 3D human pose estimation. In: Eur. Conf. Comput. Vis. pp. 69–86 (2018). https://doi.org/10.1007/978-3-030-01249-6_5
11. Huang, Y., Liu, J., Xian, K., Qiu, R.C.: PoseMamba: Monocular 3D human pose estimation with bidirectional global-local spatio-temporal state space model. In: AAAI. pp. 3842–3850 (2025). https://doi.org/10.1609/aaai.v39i4.32401
12. Ionescu, C., Papava, D., Olaru, V., Sminchisescu, C.: Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments. IEEE Trans. Pattern Anal. Mach. Intell. **36**(7), 1325–1339 (2014). https://doi.org/10.1109/TPAMI.2013.248
13. Iskakov, K., Burkov, E., Lempitsky, V., Malkov, Y.: Learnable triangulation of human pose. In: Int. Conf. Comput. Vis. pp. 7717–7726 (2019). https://doi.org/10.1109/ICCV.2019.00781
14. Jiang, J., Zhang, M., Li, J., Liu, L., Cui, Y., Ji, W., Piao, Y., Lu, H.: StructMamPose: From sequential perception to structural reasoning for 3D human pose estimation. In: Int. Conf. Mach. Learn. (2026)
15. Kang, H., Wang, Y., Liu, M., Wu, D., Liu, P., Yang, W.: Double-chain graph convolution Transformer for 3D human pose estimation. IEEE Trans. Multimedia **28**, 1148–1159 (2026). https://doi.org/10.1109/TMM.2025.3632636

16. Li, W., Liu, H., Ding, R., Liu, M., Wang, P., Yang, W.: Exploiting temporal contexts with strided Transformer for 3D human pose estimation. IEEE Trans. Multimedia **25**, 1282–1293 (2023). https://doi.org/10.1109/TMM.2022.3141231
17. Liu, J., Liu, M., Liu, H., Li, W.: TCPFormer: Learning temporal correlation with implicit pose proxy for 3D human pose estimation. In: AAAI. pp. 5478–5486 (2025). https://doi.org/10.1609/aaai.v39i5.32583
18. Liu, Y., Tian, Y., Zhao, Y., Yu, H., Xie, L., Wang, Y., Ye, Q., Jiao, J., Liu, Y.: VMamba: Visual state space model. In: Adv. Neural Inform. Process. Syst. pp. 103031–103063 (2024). https://doi.org/10.52202/079017-3273
19. von Marcard, T., Henschel, R., Black, M.J., Rosenhahn, B., Pons-Moll, G.: Recovering accurate 3D human pose in the wild using IMUs and a moving camera. In: Eur. Conf. Comput. Vis. pp. 614–631 (2018). https://doi.org/10.1007/978-3-030-01249-6_37
20. Martinez, J., Hossain, R., Romero, J., Little, J.J.: A simple yet effective baseline for 3D human pose estimation. In: Int. Conf. Comput. Vis. pp. 2640–2649 (2017). https://doi.org/10.1109/ICCV.2017.288
21. Martini, E., Boldo, M., Aldegheri, S., Valè, N., Filippetti, M., Smania, N., Bertucco, M., Picelli, A., Bombieri, N.: Enabling gait analysis in the telemedicine practice through portable and accurate 3D human pose estimation. Comput. Methods Programs Biomed. **225**, 107016 (2022). https://doi.org/10.1016/j.cmpb.2022.107016
22. Mehraban, S., Adeli, V., Taati, B.: MotionAGFormer: Enhancing 3D human pose estimation with a Transformer-GCNFormer network. In: IEEE/CVF Winter Conf. Appl. Comput. Vis. pp. 6905–6915 (2024). https://doi.org/10.1109/WACV57701.2024.00677
23. Mehta, D., Rhodin, H., Casas, D., Fua, P., Sotnychenko, O., Xu, W., Theobalt, C.: Monocular 3D human pose estimation in the wild using improved CNN supervision. In: Int. Conf. 3D Vis. (3DV). pp. 506–516 (2017). https://doi.org/10.1109/3DV.2017.00064
24. Mehta, D., Sotnychenko, O., Mueller, F., Xu, W., Elgharib, M., Fua, P., Seidel, H.P., Rhodin, H., Pons-Moll, G., Theobalt, C.: XNect: Real-time multi-person 3D motion capture with a single RGB camera. ACM Trans. Graph. **39**(4), 82:1–82:17 (2020). https://doi.org/10.1145/3386569.3392410
25. Mehta, D., Sridhar, S., Sotnychenko, O., Rhodin, H., Shafiei, M., Seidel, H.P., Xu, W., Casas, D., Theobalt, C.: VNect: Real-time 3D human pose estimation with a single RGB camera. ACM Trans. Graph. **36**(4), 44:1–44:14 (2017). https://doi.org/10.1145/3072959.3073596
26. Newell, A., Yang, K., Deng, J.: Stacked hourglass networks for human pose estimation. In: Eur. Conf. Comput. Vis. pp. 483–499 (2016). https://doi.org/10.1007/978-3-319-46484-8_29
27. Pavlakos, G., Zhou, X., Derpanis, K.G., Daniilidis, K.: Coarse-to-fine volumetric prediction for single-image 3D human pose. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 7025–7034 (2017). https://doi.org/10.1109/CVPR.2017.139
28. Pavllo, D., Feichtenhofer, C., Grangier, D., Auli, M.: 3D human pose estimation in video with temporal convolutions and semi-supervised training. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 7753–7762 (2019). https://doi.org/10.1109/CVPR.2019.00794
29. Peng, J., Zhou, Y., Mok, P.Y.: KTPFormer: Kinematics and trajectory prior knowledge-enhanced Transformer for 3D human pose estimation. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 1123–1132 (2024). https://doi.org/10.1109/CVPR52733.2024.00113

30. Qian, X., Tang, Y., Zhang, N., Han, M., Xiao, J., Huang, M.C., Lin, R.S.: HSTFormer: Hierarchical spatial-temporal Transformers for 3D human pose estimation. arXiv preprint arXiv:2301.07322 (2023). https://doi.org/10.48550/arXiv.2301.07322
31. Rhodin, H., Spörri, J., Katircioglu, I., Constantin, V., Meyer, F., Müller, E., Salzmann, M., Fua, P.: Learning monocular 3D human pose estimation from multi-view images. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 8437–8446 (2018). https://doi.org/10.1109/CVPR.2018.00880
32. Shan, W., Liu, Z., Zhang, X., Wang, S., Ma, S., Gao, W.: P-STMO: Pre-trained spatial temporal many-to-one model for 3D human pose estimation. In: Eur. Conf. Comput. Vis. pp. 461–478 (2022). https://doi.org/10.1007/978-3-031-20065-6_27
33. Tang, Z., Qiu, Z., Hao, Y., Hong, R., Yao, T.: 3D human pose estimation with spatio-temporal criss-cross attention. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 4790–4799 (2023). https://doi.org/10.1109/CVPR52729.2023.00464
34. Tekin, B., Rozantsev, A., Lepetit, V., Fua, P.: Direct prediction of 3D body poses from motion compensated sequences. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 991–1000 (2016). https://doi.org/10.1109/CVPR.2016.113
35. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: Adv. Neural Inform. Process. Syst. pp. 5998–6008 (2017)
36. Yang, Z., Bai, J., Li, H., Dai, J., Pan, J., Yin, Y., Li, B.: VIMCAN: Visual-inertial 3D human pose estimation with hybrid Mamba-cross-attention network. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 28458–28467 (2026)
37. Yu, B.X.B., Zhang, Z., Liu, Y., Zhong, S.H., Liu, Y., Chen, C.W.: GLA-GCN: Global-local adaptive graph convolutional network for 3D human pose estimation from monocular video. In: Int. Conf. Comput. Vis. pp. 8784–8795 (2023). https://doi.org/10.1109/ICCV51070.2023.00810
38. Zhang, J., Tu, Z., Yang, J., Chen, Y., Yuan, J.: MixSTE: Seq2seq mixed spatio-temporal encoder for 3D human pose estimation in video. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 13232–13242 (2022). https://doi.org/10.1109/CVPR52688.2022.01288
39. Zhang, X., Bao, Q., Cui, Q., Yang, W., Liao, Q.: Pose Magic: Efficient and temporally consistent human pose estimation with a hybrid Mamba-GCN network. In: AAAI. pp. 10248–10256 (2025). https://doi.org/10.1609/aaai.v39i10.33112
40. Zhao, L., Peng, X., Tian, Y., Kapadia, M., Metaxas, D.N.: Semantic graph convolutional networks for 3D human pose regression. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 3420–3430 (2019). https://doi.org/10.1109/CVPR.2019.00354
41. Zhao, Q., Zheng, C., Liu, M., Wang, P., Chen, C.: PoseFormerV2: Exploring frequency domain for efficient and robust 3D human pose estimation. In: IEEE Conf. Comput. Vis. Pattern Recog. pp. 8877–8886 (2023). https://doi.org/10.1109/CVPR52729.2023.00857
42. Zhu, L., Liao, B., Zhang, Q., Wang, X., Liu, W., Wang, X.: Vision Mamba: Efficient visual representation learning with bidirectional state space model. In: Int. Conf. Mach. Learn. pp. 62429–62442 (2024)
43. Zhu, W., Ma, X., Liu, Z., Liu, L., Wu, W., Wang, Y.: MotionBERT: A unified perspective on learning human motion representations. In: Int. Conf. Comput. Vis. pp. 15039–15053 (2023). https://doi.org/10.1109/ICCV51070.2023.01385